End-to-End Autonomous Drone Target Tracking using Vision

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***Abstract*—Target uncertainty, limited onboard vision, and control constraints make UAV tracking hard. I implement an end-to-end DRL framework that maps camera input directly to velocity commands, removing the traditional detection–planning–control pipeline. Using a SAC-based policy, the system achieves faster response and more robust tracking than conventional methods.**

***Index Terms*—deep reinforcement learning (DRL); end-to-end control; neural network; unmanned aerial vehicle (UAV)**

I. Introduction

Autonomous drones are increasingly being deployed in a wide range of navigation and tracking tasks, such as surveillance, search and rescue, and dynamic target following. These applications require UAVs to operate in complex and dynamic environments while maintaining real-time responsiveness and stability.

Traditional approaches to UAV control and motion planning are largely based on modular architectures, where perception, state estimation, and control are designed as separate components. Extensive work has been done in motion planning, trajectory optimization, and whole-body control to improve system performance and stability [1] [3] [4] [7]. In addition, control-theoretic methods such as hierarchical inverse dynamics and Lyapunov-based control have been widely studied to ensure robustness and stability in dynamic systems [5] [6]. While these methods provide strong theoretical guarantees, they rely heavily on accurate modeling and careful system integration.

However, such modular pipelines (e.g., detection $\rightarrow$ planning $\rightarrow$ control) suffer from several inherent limitations, including error accumulation across modules, increased latency, and reduced adaptability to rapidly changing target dynamics. Furthermore, perception and state estimation methods, such as filtering and model-based estimation, introduce additional delays and uncertainties that degrade tracking performance [2] [8].

To address these challenges, end-to-end learning-based approaches have recently gained increasing attention. By directly mapping raw sensory inputs, such as onboard camera frames, to control actions, these methods eliminate intermediate representations and significantly simplify the system architecture.

In this work, I propose an end-to-end vision-based control framework for UAV dynamic target tracking. Compared with traditional pipelines that rely on object detection and filtering modules, the proposed approach directly outputs control commands from visual inputs, resulting in reduced latency and improved tracking performance under rapidly changing target motion.

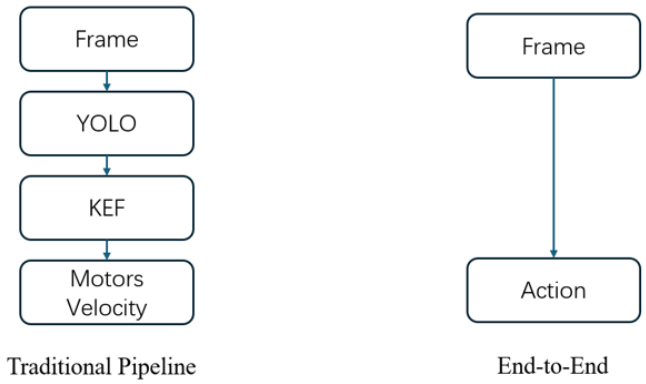


Fig. 1. Traditional pipeline vs. end-to-end approach.

II. Implementation and Results

A. Framework

Based on DRL, I design an end-to-end control framework that directly outputs velocity commands from raw camera images during UAV–target interaction. Instead of relying on manually designed features, the model learns hierarchical representations through a deep neural network that connects perception and control. The network output is denormalized into control commands and fed directly into the controller, forming a closed-loop perception–control system for dynamic target tracking.

B. Neural Network Training

The designed actor network architecture and critic network architecture for end-toend learning are shown in Figure 2. Their backbone networks are both composed of three convolution layers and spatial index normalization layers. (a) Actor network: RGB images are processed through multiple convolutional layers to extract visual features, followed by fully connected layers to output the parameters (mean and

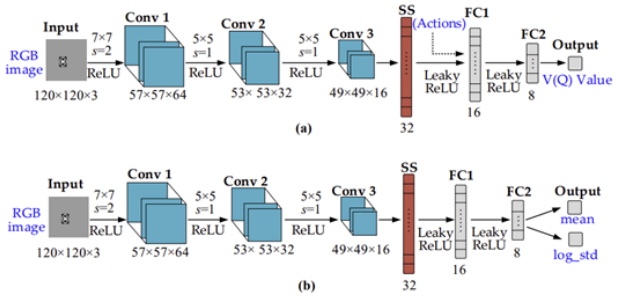


Fig. 2. Network architectures for the Soft Actor-Critic (SAC) framework.

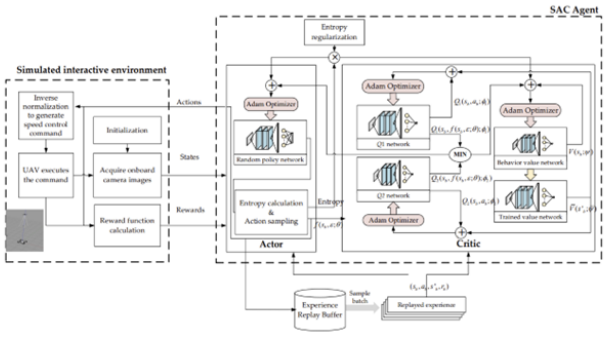


Fig. 3. End-to-end SAC-based framework for UAV target tracking, mapping visual input directly to control commands.

log standard deviation) of the action distribution. (b) Critic network: a similar convolutional backbone is used to encode image features, which are then combined with action inputs to estimate the Q-value for state–action pairs. Both networks enable end-to-end learning from visual observations to control decisions.

The reward function is composed of three components: distance-based reward r_1, direction-based reward r_2, and termination reward r_3. The term r_1 encourages the UAV to approach the target by comparing the current distance d_r with the previous distance $d_{r,\text{last}}$. Positive rewards are given when the UAV moves closer to the target and are accumulated over consecutive approaching steps, while penalties are assigned when the UAV moves away. The term r_2 evaluates the alignment between the action direction $\mathbf{a}$ and the relative direction to the target $\mathbf{a}_{\theta_r}$, computed via the angle θ_{error}. Actions aligned with the target direction are rewarded, while larger deviations result in increasing penalties.

The term r_3 handles episode termination conditions. A large negative reward is assigned if the target is lost (i.e., $d_r > d_{\text{lim}}$), while a positive reward weighted by consecutive success steps is given when the UAV reaches a region close to the target ($d_r < d_{r,\text{thresh}}$). The overall reward is defined as:

$$r = \begin{cases} r_3, & d_r > d_{\text{lim}} \\ w_1 r_1 + w_2 r_2 + r_3, & \text{otherwise} \end{cases} \tag{1}$$

where w_1 and w_2 are weighting coefficients.

To stabilize training and mitigate overestimation bias, I adopt a twin Q-network structure within the SAC framework. Specifically, two Q-functions $Q_1(s, a; \phi_1)$ and $Q_2(s, a; \phi_2)$ are maintained, and the minimum of their outputs is used for value estimation. The value network is trained by minimizing:

$$J_V(\psi) = \mathbb{E}_{s_t \sim D}\left[\frac{1}{2}\left(V_\pi(s_t;\psi) - \mathbb{E}_{a_t \sim \pi_\theta}\left[\min_{i=1,2} Q_i(s_t, a_t; \phi_i) - \alpha \log \pi(a_t|s_t;\theta)\right]\right)^2\right] \tag{2}$$

The policy network is optimized to maximize expected return while encouraging exploration:

$$J_\pi(\theta) = \mathbb{E}_{\substack{s_t \sim D \\ \epsilon_t \sim \mathcal{N}}}\Big[\log \pi(f(s_t, \epsilon_t; \theta) \mid s_t; \theta) - \min_{i=1,2} Q_i(s_t, f(s_t, \epsilon_t; \theta); \phi_i)\Big] \tag{3}$$

Based on this formulation, I implement an end-to-end SAC-based training framework that directly learns velocity control policies from visual inputs.

C. Training

During training, the agent interacts with the environment by sampling actions from the policy network and collecting transitions into the replay buffer. The SAC framework updates the value network, twin Q-networks, and policy network using stochastic gradient descent, where the minimum of the two Q-values is used to reduce overestimation bias. Entropy regularization encourages exploration by promoting stochastic policies. The target value network is updated using a soft update strategy to improve training stability. After training, the learned policy can directly map visual observations to velocity control commands for real-time UAV target tracking.

In this subsection, I train the policy network using the proposed method and evaluate its performance through simulation. To assess task completion ability, 100 trials with random initial positions are conducted, where the target is fixed at $(0, 0, 0)$.

The training setup is configured as follows. The replay buffer size is set to 10,000, and the batch size is 128. The discount factor is 0.99, and each episode is limited to 50 steps. The learning rate for all networks is 0.0003.

D. Result

Two approaches are considered in this work. Baseline: A modular pipeline consisting of YOLOv5 for object detection, a SORT tracker for target state estimation, and a PD controller for motion control. End-to-End: A deep reinforcement learning-based control framework that directly maps raw image observations to motor velocity commands.

The simulation environment is built using AirSim and RVIZ. AirSim is used for realistic UAV dynamics, image acquisition, and training of the DRL policy through environment

Algorithm 1 End-to-End SAC Training for UAV Target Tracking

Input: onboard image observation s_t
Output: velocity control command a_t

Initialize learning rates $\lambda_V, \lambda_Q, \lambda_\pi$
Initialize network parameters $\theta, \phi_1, \phi_2, \psi, \bar{\psi}$
Set entropy coefficient α and target update rate τ
Initialize replay buffer $\mathcal{D}$
while training not converged **do**
  Reset environment and obtain initial state s_0
  for each time step t **do**
    Sample action $a_t \sim \pi_\theta(\cdot|s_t)$
    Execute a_t, observe r_{t+1}, s_{t+1}
    Store $(s_t, a_t, r_{t+1}, s_{t+1})$ in $\mathcal{D}$
    Sample mini-batch $(s, a, r, s') \sim \mathcal{D}$
    Update value network:
      $J_V(\psi)$
    Update Q-networks:
      $\phi_i \leftarrow \phi_i - \lambda_Q \nabla_{\phi_i} J_Q(\phi_i),\ i = 1, 2$
    Update policy network:
      $\theta \leftarrow \theta - \lambda_\pi \nabla_\theta J_\pi(\theta)$
    Update target network:
      $\bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$
    if termination condition satisfied **then**
      break
    end if
  end for
end while

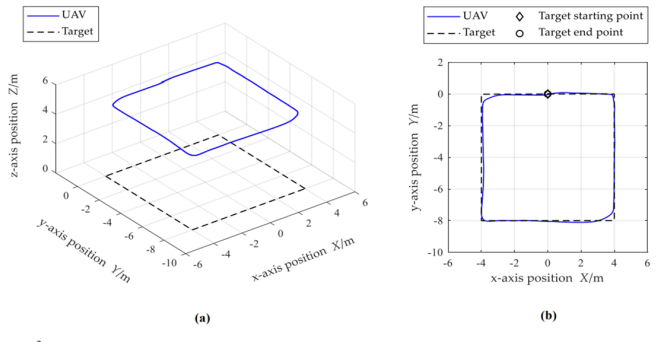


Fig. 4. UAV tracking performance on a square trajectory. (a) Three-dimensional view of the UAV (blue) and target (dashed) trajectories, showing consistent altitude tracking. (b) Top-down view, where the UAV closely follows the target path with minimal deviation. The start and end points of the target trajectory are marked.

interaction. In contrast, RVIZ is mainly used for lightweight visualization and debugging, especially for the end-to-end pipeline, where intermediate states are not explicitly available. Compared to AirSim, RVIZ allows faster iteration and clearer inspection of control outputs, making it suitable for validating policy behavior and diagnosing tracking performance during development.

The tracking performance on a square trajectory is shown in Fig.4. In the 3D view (Fig.4(a)), the UAV maintains a stable altitude while following the horizontal motion of the target. In the top-down view (Fig.4(b)), the UAV trajectory closely overlaps with the target path, with only minor deviations observed at the corners due to abrupt direction changes.

These results demonstrate that the proposed end-to-end policy can effectively handle piecewise-linear trajectories with sharp turns, maintaining both spatial accuracy and control stability. The small tracking error at corners indicates fast response but also highlights the challenge of sudden motion changes in dynamic tracking tasks.

The tracking performance of the proposed method is illustrated in Fig. 5 and Fig. 6. As shown in Fig. 5, the UAV trajectory closely follows the target trajectory with only small deviations, indicating that the learned policy can generate stable and responsive control commands in real time. Even under continuous and smooth motion, the UAV maintains consistent tracking accuracy.

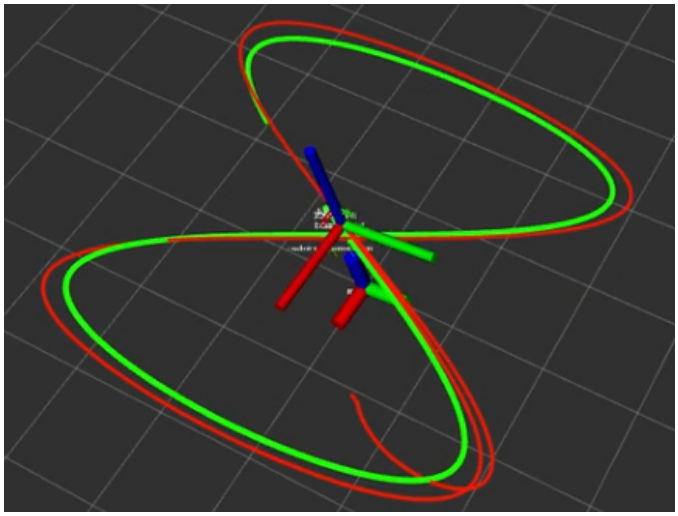


Fig. 5. UAV trajectory tracking performance in the simulation environment. The green curve represents the reference trajectory of the target, while the red curve denotes the UAV trajectory generated by the learned policy. The UAV is able to closely follow the target motion with minor deviation, demonstrating stable tracking behavior under dynamic conditions.

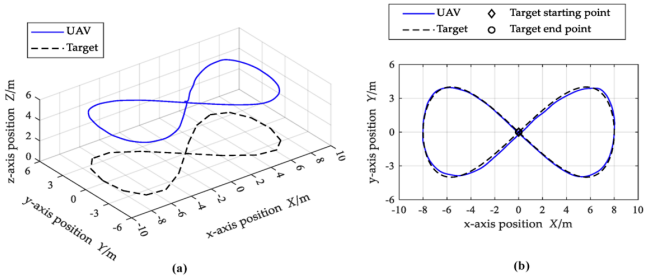


Fig. 6. UAV target tracking results under different trajectory patterns. (a) Three-dimensional tracking performance showing the UAV trajectory (blue) and target trajectory (dashed). (b) Planar projection of a figure-eight trajectory, where the UAV successfully tracks the target through sharp turns and direction changes.

Fig. 6 further demonstrates the robustness of the proposed method under more complex trajectory patterns. In the three-dimensional case shown in Fig. 6(a), the UAV is able to maintain spatial alignment with the target despite variations in altitude and direction. In the planar figure-eight trajectory shown in Fig. 6(b), the UAV successfully tracks the target through rapid direction changes and high-curvature segments. These results indicate that the end-to-end DRL framework can generalize to diverse motion patterns and maintain reliable tracking performance.

Compared with traditional modular pipelines, the proposed method reduces latency introduced by intermediate modules and enables faster response to target motion changes, which is critical for dynamic tracking tasks.

Fig.7 compares the tracking performance of the baseline modular pipeline and the proposed end-to-end method. In the position tracking results (top row), the baseline approach exhibits a clear phase lag relative to the target trajectory, particularly during segments with rapid changes in direction. This delay is mainly caused by the sequential processing of detection, tracking, and control modules, which introduces cumulative latency.

In contrast, the proposed end-to-end method significantly reduces this lag, enabling the UAV to respond more promptly to target motion. The UAV trajectory aligns more closely with the reference signal, indicating improved tracking accuracy. This improvement is more evident near turning points, where fast response is critical.

The velocity tracking results (bottom row) further highlight the difference between the two approaches. The baseline method produces noisier and less synchronized velocity outputs, with noticeable deviations from the command signals. On the other hand, the proposed method generates smoother and more consistent velocity profiles that better match the desired commands, indicating a more stable control behavior.

Overall, these results demonstrate that the end-to-end DRL framework effectively reduces perception-to-control delay, leading to improved responsiveness and more accurate dynamic target tracking compared to the traditional modular pipeline.

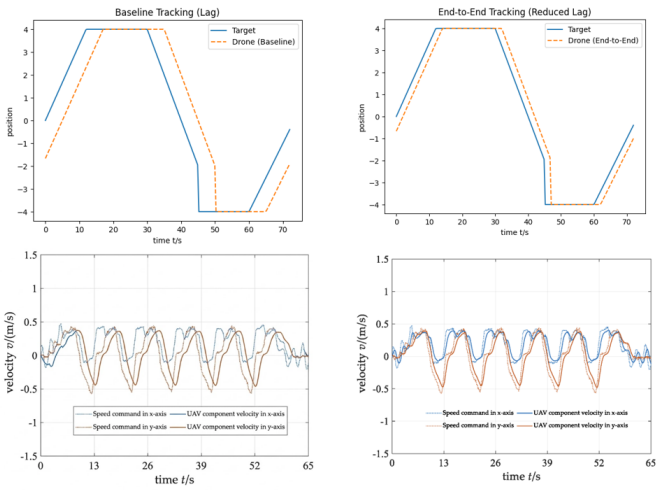


Fig. 7. Comparison between the baseline modular pipeline and the proposed end-to-end method. Left: baseline approach exhibiting significant tracking delay. Right: proposed method with reduced lag. Top: position tracking results. Bottom: velocity tracking results in both x and y directions. The baseline shows noticeable phase lag and larger tracking error, especially during rapid changes, while the proposed method achieves closer alignment with the target trajectory and smoother velocity response, demonstrating improved accuracy and responsiveness.

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